

## Enhancing T Cell Chemotaxis and Infiltration in Glioblastoma.

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Glioblastoma is an immunologically 'cold' tumor, which are characterized by absent or minimal numbers of tumor-infiltrating lymphocytes (TILs). For those tumors that have been invaded by lymphocytes, they are profoundly exhausted and ineffective. While many immunotherapy approaches seek to reinvigorate immune cells at the tumor, this requires TILs to be present. Therefore, to unleash the full potential of immunotherapy in glioblastoma, the trafficking of lymphocytes to the tumor is highly desirable. However, the process of T cell recruitment into the central nervous system (CNS) is tightly regulated. Naïve T cells may undergo an initial licensing process to enter the migratory phenotype necessary to enter the CNS. T cells then must express appropriate integrins and selectin ligands to interact with transmembrane proteins at the blood-brain barrier (BBB). Finally, they must interact with antigen-presenting cells and undergo further licensing to enter the parenchyma. These T cells must then navigate the tumor microenvironment, which is rich in immunosuppressive factors. Altered tumoral metabolism also interferes with T cell motility. In this review, we will describe these processes and their mediators, along with potential therapeutic approaches to enhance trafficking. We also discuss safety considerations for such approaches as well as potential counteragents.

## Glioblastoma microenvironment-from biology to therapy.

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Glioblastoma (GBM) is the most aggressive primary brain cancer. These tumors exhibit high intertumoral and intratumoral heterogeneity in neoplastic and nonneoplastic compartments, low lymphocyte infiltration, and high abundance of myeloid subsets that together create a highly protumorigenic immunosuppressive microenvironment. Moreover, heterogeneous GBM cells infiltrate adjacent brain tissue, remodeling the neural microenvironment to foster tumor electrochemical coupling with neurons and metabolic coupling with nonneoplastic astrocytes, thereby driving growth. Here, we review heterogeneity in the GBM microenvironment and its role in low-to-high-grade glioma transition, concluding with a discussion of the challenges of therapeutically targeting the tumor microenvironment and outlining future research opportunities.

## Effector T-cell infiltration positively impacts survival of glioblastoma patients and is impaired by tumor-derived TGF- $\beta$ .

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In glioma-in contrast to various other cancers-the impact of T-lymphocytes on clinical outcome is not clear. We investigated the clinical relevance and regulation of T-cell infiltration in glioma. T-cell subpopulations from entire sections of 93 WHO°II-IV gliomas were computationally identified using markers CD3, CD8, and Foxp3; survival analysis was then done on primary glioblastomas (pGBM). Endothelial cells expressing cellular adhesion molecules (CAM) were similarly computationally quantified from the same glioma tissues. Influence of prominent cytokines (as measured by ELISA from 53 WHO°II-IV glioma lysates) on CAM-expression in GBM-isolated endothelial cells was determined using flow cytometry. The functional relevance of the cytokine-mediated CAM regulation was tested in a

transmigration assay using GBM-derived endothelial cells and autologous T-cells. Infiltration of all T-cell subsets increased in high-grade tumors. Most strikingly, within pGBM, elevated numbers of intratumoral effector T cells (T<sub>eff</sub>), cytotoxic and helper) significantly correlated with a better survival; regulatory T cells were infrequently present and not associated with GBM patient outcome. Interestingly, increased infiltration of T<sub>eff</sub> cells was related to the expression of ICAM-1 on the vessel surface. Transmigration of autologous T cells in vitro was markedly reduced in the presence of CAM-blocking antibodies. We found that TGF- $\beta$  molecules impeded transmigration and downregulated CAM-expression on GBM-isolated endothelial cells; blocking TGF- $\beta$  receptor signaling increased transmigration. This study provides comprehensive and novel insights into occurrence and regulation of T-cell infiltration in glioma. Specifically, targeting TGF- $\beta$ 1 and TGF- $\beta$ 2 might improve intratumoral T-cell infiltration and thus enhance effectiveness of immunotherapeutic approaches.

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## Lymphocyte homing and recirculation with tumor tertiary lymphoid structure formation: predictions for successful cancer immunotherapy.

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The capacity of lymphocytes continuously home to lymphoid structures is remarkable for cancer immunosurveillance and immunotherapy. Lymphocyte homing and recirculation within the tumor microenvironment (TME) are now understood to be adaptive processes that are regulated by specialized cytokines and adhesion molecule signaling cascades. Restricted lymphocyte infiltration and recirculation have emerged as key mechanisms contributing to poor responses in cancer immunotherapies like chimeric antigen receptor (CAR)-T cell therapy and immune checkpoint blockades (ICBs). Uncovering the kinetics of lymphocytes in tumor infiltration and circulation is crucial for improving immunotherapies. In this review, we discuss the current insights into the adhesive and migrative molecules involved in lymphocyte homing and transmigration. The potential mechanisms within the TME that restrain lymphocyte infiltration are also summarized. Advanced on these, we outline the determinates for tertiary lymphoid structures (TLSs) formation within tumors, placing high expectations on the prognostic values of TLSs as therapeutic targets in malignancies.

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## Analysis of PD-L1 and CD3 Expression in Glioblastoma Patients and Correlation with Outcome: A Single Center Report.

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With the advent of immunotherapies, the field of cancer therapy has been revived with new hope, especially for cancers with dismal prognoses, such as the glioblastoma multiforme (GBM). Currently, immunotherapies should potentiate the host's own antitumor immune response against cancer cells, but it has been documented that they are effective only in small subsets of patients. Therefore, accurate predictors of response are urgently needed to identify who will benefit from immune-modulatory therapies. Brain tumors are challenging in terms of treatments. The immune response in the brain is highly regulated, and the immune microenvironment in brain metastases is active with a high density of tumor-infiltrating lymphocytes (TILs, CD3+ T cells) in certain patients and, therefore, may serve as a potential treatment target. In our study, we performed immunohistochemistry for CD3 and PD-L1 along the routine assessment of the O6-methylguanine-methyltransferase (MGMT) promoter methylation status

and the IDH1 and 2 status in a single center cohort of 69 patients with GBM (58 primary tumors and 11 recurrences) who underwent standard multimodal therapies (surgery/radiotherapy/adjunctive temozolamide). We analyzed the association of PD-L1 tumor expression and TILs with overall survival (OS). The PD-L1 expression was observed in 25 of 58 (43%) newly diagnosed primary glioblastoma specimens. The sparse-to-moderate density of TILs, identified with CD3+ expression, was found in 48 of 58 (83%) specimens. Neither PD-L1 expression nor TILs were associated with overall survival. In conclusion, TILs and/or PD-L1 expression are detectable in the majority of glioblastoma samples, and even if they slightly relate to the outcome, they do not show a statistically significant correlation.

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## Magnesium-Phenolic Nanoeditor Refining Gliomatous T Cells for Metalloimmunotherapy.

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More than the sparse infiltration in glioblastoma, cytotoxic T lymphocytes (CTLs) also function inefficiently and overexpress the inhibitory markers, especially the identified NK cell receptor (NK1.1). However, most studies solely focus on how to augment tumor-infiltrating CTLs and overlook their killing maintenance. Metalloimmunotherapy has been proven to improve the functionalities of CTLs, but it has barely adapted to glioblastoma due to the severe limitations of safe delivery and the brain's physiological properties. Herein, we synthesized an amphipathic polyethylene glycol (PEG) polymer (designated as MPP) modified with the choline analogue 2-methacryloyloxyethyl phosphorylcholine (MPC) and polyphenol moieties to customize a nanoeditor (Mg2+@MK-8931@MPP) by coordinating Mg2+ and entrapping the hydrophobic BACE1 inhibitor MK-8931, then precisely redressing the gliomatous CTL sparsity and cytotoxic dysfunction. Upon MPC-assisted local accumulation in glioblastoma, Mg2+@MK-8931@MPP nanoeditors release MK-8931 to repolarize M2-like macrophages, facilitating CTL infiltration quantitatively. The xenogenetic immune adjuvant Mg2+ ultimately fortifies the T-cell receptor downstream signals to enhance the functionality of the incoming CTLs in quality, leading to the secretion of high-level antitumor cytokines and cytotoxic proteins. Further blocking the inhibitory NK1.1 on CTLs by anti-NK1.1 antibodies can extend their cytolytic endgame. Studies on T-cell-deficient and wild-type mouse models support the immunomodulating feasibility of Mg2+@MK-8931@MPP. This gliomatous CTL-tailored strategy concurrently broadens metalloimmunotherapy to glioblastoma treatment and highlights the necessity of enforcing gliomatous CTLs' functionality.

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## Inflammation and lymphocyte infiltration are associated with shorter survival in patients with high-grade glioma.

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Glioma represents a serious health burden in terms of morbidity and mortality. The prognostic significance of the lymphoid and myeloid infiltrates in glioma is not clearly determined. Moreover, the characterization of different leukocyte subsets in the tumor microenvironment relies mainly on immunohistochemistry observations, and data about their association with prognosis are contradictory. Here, we performed a comprehensive study of both the tumor-infiltrating and circulating immune compartments of patients with high-grade glioma. Nineteen tumor biopsies and 30 PBMC samples were analyzed by RNA sequencing. Validation was performed on The Cancer Genome Atlas (TCGA) RNA sequencing data from glioma and on additional 39 tumor biopsies analyzed by flow cytometry. We

identified prognostic tumor and peripheral immune signatures, which associate increased inflammation, immune infiltration and activation with shorter overall survival in high-grade glioma patients. Importantly, we confirmed our observations by flow cytometry analysis and validated the tumor-signature using the TCGA dataset. In addition, both tumor genotype and grade associated with the degree of glioma immune infiltration. Unlike in the majority of cancers, lymphocyte infiltration at the tumor site is a negative prognostic factor in glioma, suggesting the ambivalent pro-tumorigenic role of immune responses in glioma.

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## Programmed death ligand 1 expression and tumor-infiltrating lymphocytes in glioblastoma.

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Immune checkpoint inhibitors targeting programmed cell death 1 (PD1) or its ligand (PD-L1) showed activity in several cancer types. We performed immunohistochemistry for CD3, CD8, CD20, HLA-DR, phosphatase and tensin homolog (PTEN), PD-1, and PD-L1 and pyrosequencing for assessment of the O6-methylguanine-methyltransferase (MGMT) promoter methylation status in 135 glioblastoma specimens (117 initial resection, 18 first local recurrence). PD-L1 gene expression was analyzed in 446 cases from The Cancer Genome Atlas. Diffuse/fibrillary PD-L1 expression of variable extent, with or without interspersed epithelioid tumor cells with membranous PD-L1 expression, was observed in 103 of 117 (88.0%) newly diagnosed and 13 of 18 (72.2%) recurrent glioblastoma specimens. Sparse-to-moderate density of tumor-infiltrating lymphocytes (TILs) was found in 85 of 117 (72.6%) specimens (CD3+ 78/117, 66.7%; CD8+ 52/117, 44.4%; CD20+ 27/117, 23.1%; PD1+ 34/117, 29.1%). PD1+ TIL density correlated positively with CD3+ ( $P < .001$ ), CD8+ ( $P < .001$ ), CD20+ TIL density ( $P < .001$ ), and PTEN expression ( $P = .035$ ). Enrichment of specimens with low PD-L1 gene expression levels was observed in the proneural and G-CIMP glioblastoma subtypes and in specimens with high PD-L1 gene expression in the mesenchymal subtype ( $P = 5.966e-10$ ). No significant differences in PD-L1 expression or TIL density between initial and recurrent glioblastoma specimens or correlation of PD-L1 expression or TIL density with patient age or outcome were evident. TILs and PD-L1 expression are detectable in the majority of glioblastoma samples but are not related to outcome. Because the target is present, a clinical study with specific immune checkpoint inhibitors seems to be warranted in glioblastoma.

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## Revealing and Harnessing Tumour-Associated Microglia/Macrophage Heterogeneity in Glioblastoma.

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**Abstract:** Cancer heterogeneity and progression are subject to complex interactions between neoplastic cells and their microenvironment, including the immune system. Although glioblastomas (GBMs) are classified as 'cold tumours' with very little lymphocyte infiltration, they can contain up to 30-40% of tumour-associated macrophages, reported to contribute to a supportive microenvironment that facilitates tumour proliferation, survival and migration. In GBM, tumour-associated macrophages comprise either resident parenchymal microglia, perivascular macrophages or peripheral monocyte-derived cells. They are recruited by GBMs and in turn release growth factors and cytokines that affect the tumour. Notably, tumour-associated microglia/macrophages (TAMs) acquire different expression programs, which shape

the tumour microenvironment and contribute to GBM molecular subtyping. Further, emerging evidence highlights that TAM programs may adapt to specific tumour features and landscapes. Here, we review key evidence describing TAM transcriptional and functional heterogeneity in GBM. We propose that unravelling the intricate complexity and diversity of the myeloid compartment as well as understanding how different TAM subsets may affect tumour progression will possibly pave the way to new immune therapeutic avenues for GBM patients.

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## Low tumour-infiltrating lymphocyte density in primary and recurrent glioblastoma.

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Immunotherapies targeting tumour-infiltrating lymphocytes (TILs) that express the immune checkpoint molecule programmed cell death-1 (PD-1) have shown promise in preclinical glioblastoma models but have had limited success in clinical trials. To assess when glioblastoma is most likely to benefit from immune checkpoint inhibitors we determined the density of TILs in primary and recurrent glioblastoma. Thirteen cases of matched primary and recurrent glioblastoma tissue were immunohistochemically labelled for CD3, CD8, CD4 and PD-1, and TIL density assessed. CD3+ TILs were observed in all cases, with the majority of both primary (69.2%) and recurrent (61.5%) tumours having low density of TILs present. CD8+ TILs were observed at higher densities than CD4+ TILs in both tumour groups. PD-1+ TILs were sparse and present in only 25% of primary and 50% of recurrent tumours. Quantitative analysis of TILs demonstrated significantly higher CD8+ TIL density at recurrence ( $p = 0.040$ ). No difference was observed in CD3+ ( $p = 0.191$ ), CD4+ ( $p = 0.607$ ) and PD-1+ ( $p = 0.070$ ) TIL density between primary and recurrent groups. This study shows that TILs are present at low densities in both primary and recurrent glioblastoma. Furthermore, PD-1+ TILs were frequently absent, which may provide evidence as to why anti-PD-1 immunotherapy trials have been largely unsuccessful in glioblastoma.

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